Response to the reviewer

Manuscript 3400076, "Internal Camera-based Reconstruction and Closed-loop Control of Soft Robotic Arms"

Dear Dr. Murray,

We thank you and the reviewer for the careful evaluation of our manuscript. We have revised the manuscript and the Supporting Information according to all comments. Below, the comments are reproduced in italics and followed by our response; changes in the manuscript are highlighted.

Response to the Editor

Editor's requirements

> *Please ensure that the title of your manuscript is succinct, descriptive, and accurate [...] Reviewers may have suggested references [...] define all acronyms when first mentioned [...] check that all parts, insets, and footnotes of Figures and Schemes are mentioned in the caption. [...] obtain copyright permission information for any images that have not been created originally for this manuscript [...] full article titles, and optionally, DOIs.*

Title. We kept the title, which states the method (internal-camera reconstruction and closed-loop control) and the system (soft robotic arms).

Suggested references. We added the references suggested by the reviewer (Comment 2).

Acronyms. We proofread the manuscript and define every acronym at its first use in the main text and in the SI, including those that appear only in figures (e.g., BW, FE, and FC in Fig. 1, and CL and OL in Figs. 5 and 8). The title and abstract contain no undefined acronyms.

Figures and captions. We checked the text and axis labels of all figures and revised the captions so that every panel and inset is described (Comment 10).

Copyright. All figures were created by the authors for this manuscript; no reprinted or adapted material, logos, or trademark images are included.

References. The reference list gives full article titles, and we added DOIs where available.

Response to Reviewer 1

Overall assessment

> *I recommend major revision. The claimed contributions (internal-camera proprioception, learned inverse kinematics, closed-loop soft-arm control, and redundancy resolution) have each been extensively addressed in prior work. [...] Several central claims also extend beyond the evidence presented. [...]*

We thank the reviewer for the detailed and constructive assessment. We agree that the contribution lies in the integration of distributed internal cameras, markerless at inference, with model-based and learned redundancy-aware control on a multi-section arm, and we have revised the manuscript to state this precisely and to align the claims with the evidence. In summary: (i) we added the suggested references, a comparison paragraph in the Introduction, and a comparison table in the SI; (ii) we state explicitly that the curvature reconstruction does not enter the control laws and explain the encoder offset in Algorithm 1; (iii) we narrowed the scalability claim to networks retrained

with a transferred backbone, validated for one and three sections; (iv) we describe the multi-section curvature result as a model-constrained reconstruction and state its assumptions; (v) we added an offline ablation of the secondary objectives of both controllers; (vi) we quantified the coverage of the control evaluation and limited the claims on repeatability and disturbances; (vii) we documented the data split and its effect, and added a reproducibility table; and (viii) we revised the figures and captions as requested. Our responses to each comment follow; changes in the manuscript are highlighted.

Comment 1

> *1. Clarify the role of visual sensing in closed-loop control. [...] Algorithm 1 also uses a PCC inverse calculation to initialize tendon lengths from section-tip positions. [...] the resulting estimate appears to be replaced by the motor-encoder measurement at the first control iteration, before it affects the control command. The authors should explain this initialization or remove it. The abstract, contribution list, Fig. 4c, and conclusion should clearly distinguish [...]*

We thank the reviewer for this careful reading. The reviewer is correct that the curvature reconstruction does not enter either control law: both controllers close the loop on the tip pose estimated from the camera images, the model-based controller additionally uses the motor encoders to evaluate the PCC Jacobian and the null-space term, and the curvature estimate is evaluated as a separate sensing capability. We now state this explicitly in the Abstract, the contribution list, Section 3.3, the caption of Fig. 4c (which now shows the encoder path), and the Conclusion.

The initial PCC estimate in Algorithm 1 is not discarded. The tendon lengths $\mathbf{l}_0$ obtained from the camera-estimated section-tip positions through the inverse PCC map are used once, before the loop, to set the encoder offset $\mathbf{l}_{\text{off}} = \mathbf{l}_{\text{enc}} - \mathbf{l}_0$, so that the encoder readings inside the loop are expressed in the PCC tendon-length frame used by $\mathbf{J}(\mathbf{l})$ and by the null-space term $\mathbf{N}(\bar{\mathbf{l}} - \mathbf{l})$. Algorithm 1 and Section 3.3.2 now show this step explicitly.

Comment 2

> *2. Establish originality claims through direct comparison with prior work. [...] The authors should explain what each close prior study established and identify the specific capability added here. The claim that no prior framework combines comparable sensing with closed-loop null-space control is too broad without this analysis.*

We thank the reviewer for these references. We added Hofer et al., Thuruthel et al., Han et al., and Lu et al. (Yoo et al. was already cited as Ref. [15]) and a paragraph in the Introduction that states what each study established, together with a comparison table in the SI (Table S10). Hofer et al. obtain a higher orientation accuracy (about 1° at 30 Hz) than our single-section estimate (3.11° at about 8 Hz), on a two-angle spherical joint with an internal painted pattern and illumination; our system estimates the full 6-DoF tip pose of an extensible section without internal markers and is applied to three sections. Thuruthel et al. learned closed-loop inverse kinematics with end-effector feedback, Han et al. and Yoo et al. estimate shape from internal cameras without closing a control loop, and Lu et al. servo the shape with fiber Bragg grating sensing. The capability we add is the combination of distributed internal cameras on a multi-section arm with closed-loop control that explicitly exploits the actuation redundancy, through model-based null-space control and through a learned inverse kinematics with a secondary objective. We narrowed the novelty statement in the Introduction and the contribution list to this combination.

Comment 3

> *3. Substantiate or narrow the scalability claim. The estimator is not invariant to the number of arm sections. [...] The statement that only the first layer must be adjusted also conflicts with Fig. 3 because the regression head changes as well.*

We agree. The input layer and the regression head both depend on the number of sections, and separate networks were trained for the one- and three-section arms. We corrected the statement in Section 3.1 and rephrased the Introduction, the contribution list, and the Conclusion accordingly: the backbone is transferred and the network is retrained, and only $n = 1$ and $n = 3$ were validated. We also removed the word "scalable" where it implied that no retraining is needed.

Comment 4

> *4. Strengthen the ground truth for full-body reconstruction. [...] Fig. S2b does not provide sufficient validation of N_C [...] Otherwise, the result should be described as model-constrained reconstruction, with its assumptions stated explicitly, including planar bending within each section, no torsion, and no arc-dependent bending direction.*

We agree. For the multi-section arm, inter-section marker occlusion prevented tracking all backbone markers during data collection, and the curvature labels were obtained from the motion-captured section-tip poses through N_C. We therefore now describe the multi-section result as a model-constrained reconstruction and state its assumptions explicitly in Section 2.2 and Section 4.2.2: planar bending within each section, no torsion, and a bending direction that is constant along each section. The errors in Fig. 7b quantify the agreement with these model-constrained labels. For Fig. S2b, we now report the number of validation configurations (20, posed manually with all subsection markers visible and recorded in a separate session), that the error bars are one standard deviation across configurations, and that these configurations are independent of the data used to train N_C (single-section dataset) and the three-section curvature estimator. We also report the error of N_C on held-out single-section test data ($N = 500$): 0.38–0.43 mm at $L/3$, 1.20–1.37 mm at $2L/3$, and 2.49–2.94 mm at L for the three curvature models (SI Section 4.2).

Comment 5

> *5. Correct the curvature-coordinate inconsistency. [...] The paper should consistently use either normalized arc length $u = x/L$ or dimensional arc length $x \in [0, L]$, and revise Eqs. 4–6, SI Eqs. S3–S9, the translation integral, nomenclature, and coefficient units accordingly.*

We thank the reviewer. The implementation and the SI equations use the dimensional arc length $x \in [0, L]$. We corrected the main text accordingly (Eqs. 4–6, the translation integral without the factor L, and the evaluation points $x = L/3, 2L/3, L$), the initialization sentence in SI Section 2.2, and the nomenclature in Table S1, which now gives the units of c_0, c_1, and c_2 (mm^{-1}, mm^{-2}, mm^{-3}).

Comment 6

> *6. Quantitatively evaluate the secondary objective. [...] The objective should therefore might described as actuator-range centering unless effort is measured. For both controllers, an ablation with the secondary objective disabled using identical targets would be helpful. [...]*

We agree that $H(\mathbf{l})$ centers the tendon lengths in their actuation range and does not measure effort, which was not recorded. We now describe it as actuator-range centering and removed the statements about effort (Introduction, Section 3.3.2, Conclusion). To isolate its effect without new hardware experiments, we performed offline ablations on identical targets for both controllers (new SI Section 5.3, Fig. S8, and Table S13). For the model-based controller, replaying the control law on the PCC model for the 20 evaluation targets gives a final position error of 5.25 ± 2.23 mm with and 5.84 ± 1.85 mm without the null-space term, while the RMS deviation of the tendon lengths from $\bar{\mathbf{l}}$ decreases from 12.00 ± 3.18 mm to 7.48 ± 1.96 mm ($H(\mathbf{l})$: 693 ± 396 vs. 269 ± 135). For the data-driven controller, we retrained the inverse-kinematics network with and without the secondary term ($\zeta = 0.01$ and $\zeta = 0$, same data, split, and seed) and evaluated it through the frozen forward network on the same targets: the secondary term reduces $H(\mathbf{l})$ from 487 ± 257 to 371 ± 166 (RMS tendon deviation from 10.1 to 8.8 mm), at the cost of a small increase of the position error predicted by the forward network (5.1 ± 2.1 vs. 4.6 ± 2.3 mm). Unlike the null-space projection, the penalty in the training loss trades some primary-task accuracy for range centering, which we now state in Section 3.3.2. These results are simulated through the respective models and are reported as such.

Comment 7

> *7. Expand and better characterize the control evaluation. [. . .] The authors should report the target coordinates, associated bending and compression ranges, and the fraction of the sampled workspace represented by the evaluation set. Each of the 20 targets is visited only once. [. . .] Either additional load magnitudes and directions should be tested or the claims should be limited.*

We thank the reviewer. We replaced the bending range in Section 3.2 with the measured one (single section: maximum 57°, 95th percentile 36°; multi-section 95th percentiles of 30°, 32°, and 34°), and we added the coordinates of all targets with their bending angle and length change (Tables S8 and S9) and the coverage of the evaluation set (SI Section 6.3 and Fig. S9). The single-section targets and trajectories reach 33.7° and 40.9°, close to the 95th percentile of the training data (35.6°); the 20 targets span 9.2% (single) and 16.4% (multi) of the convex hull of the sampled positions. We revised the text to state that the reported standard deviations describe the variation across targets visited once, not repeatability, and that trajectory waypoints are serially correlated. The disturbance experiment is now described as the compensation of a single constant 50 g off-center load rather than general robustness. Repeated trials, non-planar and boundary trajectories, and further load conditions are listed as limitations and future work in Section 5.

Comment 8

> *8. Establish independence of training and evaluation data. [. . .] The authors should state: how samples were randomized; whether test data were separated by session or configuration neighborhood; whether controller targets and trajectories were excluded from training; and whether experiments span different days, camera settings, coverings, or robot builds.*

We added this information to Section 3.1.1 and SI Section 4.1. All samples of each dataset were pooled and randomly permuted with a fixed seed before the 85/10/5 division; the split was not blocked by session or by configuration neighborhood. Because each commanded configuration of the multi-section arm was held for about 3 s and recorded at 1 Hz, some test samples share their configuration with training samples. To quantify the effect, we evaluated the test error with and without these samples: for the multi-section arm, the tip position errors become 4.60, 14.72, and 27.99 mm (instead of 4.44, 14.01, and 26.45 mm) (Table S14), and for the single-section arm,

samples close to a training configuration are not more accurate than the others (4.08 vs. 3.86 mm). Session-blocked evaluation is listed as a limitation. The control targets were selected among the sampled workspace configurations and were therefore not excluded from training; the closed-loop experiments use new camera images, and the trajectories are not part of the datasets. The single-section data were collected in one day and the multi-section data on two days, with the same camera settings and one build of each arm **[CONFIRM]**; the two arms use different coverings (corrugated silicone and stretchable fabric, Section 2.1).

Comment 9

> *9. Provide the missing reproducibility details. [. . .] The SI is a good place for a single reproducibility table containing these settings. Claims of robustness to lighting or hardware variation should be limited unless such variation was explicitly tested.*

We added a reproducibility table to the SI (Table S11) that collects the camera and acquisition settings, the network architectures, the training hyperparameters (optimizer, learning-rate schedule, maximum epochs, early stopping, batch size, random seed, normalization, angle handling, augmentation) for all networks including N_C, N_p^3, and N_l^3, the actuation hardware and pretensioning procedure, the tendon material and diameter, the sampled tendon ranges, and the acquisition sessions (Table S12). We also limited the statements on lighting: image augmentations were used during training, but robustness to lighting or hardware variations was not tested, and this is now stated in Section 5.

Comment 10

> *10. Revise the figures and captions. Several figures—particularly Figs. 1B, 2, 3, and 5—are difficult to interpret. [. . .] The paper should use a consistent flow direction, label all network inputs and outputs, and distinguish clearly between the networks used for pose estimation, curvature reconstruction, forward kinematics, and inverse kinematics.*

We revised the figures and captions. Figure 3 has been redrawn with a single left-to-right flow, the name of each network on its path, the inputs and outputs of every network with tensor dimensions, and colors that separate pose estimation, curvature reconstruction, forward kinematics, and inverse kinematics; the caption now lists the inputs and outputs of all networks and states which architecture is used for the single- and multi-section kinematic networks. The captions of Figs. 1 and 2 now describe every labeled element (including the boundary arcs L_1–L_4, which are geometric constructions and not tendons), and Fig. 5b now uses equal axis scaling.

Minor comments

Comment 11

> *Use 'markerless at measured inferences' rather than simply 'markerless.' External optical markers are required to generate the supervised labels, even though markers are not required during deployment.*

We now write "markerless at inference" and explain at first use that external motion-capture markers are used only to generate the training labels.

Comment 12

> *Use either 'section' or 'segment' consistently throughout the manuscript, figures, algorithms, and SI.*

We now use "section" throughout the manuscript, figures, algorithms, and SI.

Comment 13

> *The calibration paragraph says that 'all ten tendons' are tightened, but the multi-section arm has nine tendons and the single-section arm has three. Correct the number and state whether the same preload procedure is used for both systems.*

We now state the number of tendons explicitly (three for the single-section arm and nine for the multi-section arm) and that the same pretensioning procedure is used for both arms, including its criterion and the motor current limit (Section 2.1 and Table S11).

Comment 14

> *Identify the camera manufacturer and model, lens or field of view, nominal frame rate, and the physical size and mass of each camera and mount.*

The camera manufacturer and model, lens and field of view, nominal frame rate, and the size and mass of the camera and mount are now given in Section 2.1 and in Table S11.

Comment 15

> *The claim that images are acquired simultaneously should be supported by the synchronization method [. . .]*

The acquisition is synchronized in software only: each camera is read by its own capture thread, and the controller uses the latest frame of each camera. We removed the word "simultaneously" and describe this mechanism in Section 2.1 and in the caption of Fig. 1.

Comment 16

> *Explain how angular sine/cosine outputs are converted back to angles and how losses for position, orientation, length, and curvature are balanced after normalization.*

The networks regress the angles directly; the sine and cosine of the predicted and target angles are only used inside the loss to remove the 2π discontinuity, so no conversion back to angles is needed. Positions, lengths, and curvature coefficients are standardized with the mean and standard deviation of the training set, and all loss terms have equal weight. This is now stated in SI Section 3.1 and Table S11.

Comment 17

> *Clarify the negative sign in Eq. S11 and clarify how the highlighted pixels are the regions used for prediction.*

In Eq. S11 the Grad-CAM score is the negative loss, so positive weights correspond to feature-map channels whose activation reduces the prediction loss; after the ReLU of Eq. S12, the heatmap retains only these regions. The maps are normalized to $[0, 1]$ per image and shown without thresholding, so the color scale in Fig. 4a shows this relative contribution. We clarified both points in SI Section 3.2 and in the caption of Fig. 4.

Comment 18

> *In Algorithm 1, the initial line that estimates tendon lengths from intermediate positions via PCC appears unused because tendon lengths are read from encoders inside the loop. Remove it or explain its role.*

Please see our response to Comment 1: the PCC estimate sets the encoder offset used inside the loop, and Algorithm 1 now shows this step explicitly.

Comment 19

> *Use consistent Jacobian notation. J_t2l, J_l2p, and their learned representations currently alternate between a forward Jacobian, an inverse Jacobian, and a derivative of an inverse network. [. . .]*

We now use a single notation, defined in Section 3.3: $\mathbf{J}_{a\to b} = \partial b/\partial a$. The analytical PCC Jacobians are forward Jacobians ($\mathbf{J}_{l\to t}$, $\mathbf{J}_{l\to p}$) with damped pseudoinverse $\mathbf{J}^{\#}$, and the Jacobians of the learned inverse kinematics are denoted with a hat ($\hat{\mathbf{J}}_{t\to l}$, $\hat{\mathbf{J}}_{p\to l}$). The notation is used consistently in Eqs. 7–10, Algorithms 1–2, Fig. S3, and Table S2.

Comment 20

> *State whether the open-loop baselines use the same measured initial pose, planner, gains, stopping limits, and commanded waypoint sequence as the closed-loop methods. 'Open-loop' and 'feedforward' should be used consistently.*

Section 3.3.1 now states that the open-loop baselines use the same planner, waypoint spacing, damping, and target sequence as the closed-loop controllers. Their initial state is measured once by motion capture and then propagated with the model-based or data-driven kinematics instead of the camera estimate, and they execute the full waypoint sequence without a convergence check. We use "open-loop" throughout.

Comment 21

> *State the denominator used for every percentage error. [. . .]*

We now state the normalization of every percentage at its first use: the section length (150 mm) for the single-section arm; for the multi-section arm, the length from the base to the evaluated tip (150, 300, and 450 mm) for the pose estimation, and the 450 mm arm length for the control experiments.

Comment 22

> *Figure 5b intentionally uses nonuniform axis scaling. This visually distorts the reconstructed shapes. Use equal spatial scaling where possible or add a second undistorted view and explicit scale bars.*

We agree that the non-uniform scaling distorted the shapes. The shape panels of Fig. 5b now use equal axis scaling, and the caption no longer refers to non-uniform scaling.

Comment 23

> *Figure 7 alternates among 'cumulative 3D error,' 'sum of 3D errors,' and 'average 3D error for each point.' Define the statistic exactly. [. . .] Per-point RMSE or error versus normalized arc length would be clearer.*

We agree that the summed statistic was ambiguous and not comparable between the two arms. Figure 7 now shows the 3D error at each reference point against the normalized arc length, for both arms on the same horizontal axis (median and interquartile range over the 500 test samples), and the caption and Sections 4.1.2 and 4.2.2 define the statistic.

Comment 24

> *In Table 1, the text states that orientation error remains below 9 degrees, but several standard deviations are larger than the corresponding means. Make clear that only the mean is below 9 degrees and report median, 95th percentile, and maximum error.*

Table 1 now also reports the median, 95th percentile, and maximum error, and the text states that the mean orientation error is below 9° (the median is below 7°). The large standard deviations and maxima are caused by 10 of the 500 test samples (2%), most of which have motion-capture orientation labels inconsistent with the sampled range; we discuss them in SI Section 6.1 (see also our response to the comment on Table S6).

Comment 25

> *The claim that bent configurations improve accuracy is plausible but currently qualitative. The authors should plot error against bending angle before drawing this conclusion.*

We plotted the error against the bending angle (new Fig. S7). For the single-section arm, the error decreases with bending (Spearman $\rho = -0.22$, $p < 0.001$), which supports the statement in Section 4.1.1, now given with this value. For the multi-section arm, no clear dependence was found, and we revised the corresponding sentence in Section 4.2.1.

Comment 26

> *For Figs. 6, 9, and 10, identify the start direction, waypoint order, number of completed cycles, and whether the plotted paths are one run or an average. Add quantitative path errors in the caption or an accompanying table.*

The captions of Figs. 6, 9, and 10 now state that each path is a single run of one counter-clockwise cycle starting at $(r, 0)$, and a new table (Table S7) reports the radial and height RMS deviation from the reference circle for every controller, arm, and load condition. For example, under the 50 g payload the radial RMS error on the 120 mm circle is 11.46 mm in closed loop and 27.10 mm in open loop.

Comment 27

> *The discussion should distinguish demonstrated capabilities from proposed applications. Obstacle avoidance, whole-body grasping, surgical guidance, and force or contact sensing are future directions and were not evaluated here.*

We revised Section 5 to separate the capabilities evaluated in this work from future applications. Obstacle avoidance, whole-body grasping, surgical and inspection applications, and force or contact sensing are now presented explicitly as future directions that were not evaluated here.

Supporting Information

Comment 28 (SI)

> *Fig. S2b should report the numerical N_C error for each curvature model, the number of validation configurations, and whether the bars show standard deviation, standard error, or another interval.*

The caption of Fig. S2 and SI Section 4.2 now report the number of validation configurations (20), how they were obtained (posed manually with all markers visible, in a separate session not used for training), and that the error bars are one standard deviation across configurations. We also report the error of N_C for each curvature model on held-out single-section test data (please see our response to Comment 4).

Comment 29 (SI)

> *Table S6 contains highly skewed errors, particularly 11.92 +/- 25.10 mm for Section II and orientation standard deviations near 16 degrees. Report robust distribution summaries and discuss the outliers rather than relying only on mean and standard deviation.*

We added the median, interquartile range, 95th percentile, and maximum to Table S6 (and Table 1). The large standard deviations are caused by 10 of the 500 test samples (2%): nine of them are three configurations recorded three times each, whose motion-capture orientation labels imply section bending angles of 87°–174°, far outside the sampled range, which is consistent with orientation flips of the tracked rigid bodies. Excluding these samples, the Section II relative position error is 9.22 ± 4.68 mm instead of 11.92 ± 25.10 mm. We discuss these outliers in SI Section 6.1 and report both summaries.

Comment 30 (SI)

> *Figs. S4-S6 should use matched target identifiers or connecting pairs so the reader can compare the same workspace locations across controllers. Use consistent color limits when the goal is direct visual comparison.*

Figures S4–S6 now label each of the 20 targets with the same identifier in every panel, and use a single color scale per quantity within each figure. The target coordinates are listed with the same identifiers in Tables S8 and S9.

Comment 31 (SI)

> *Tables S3 and S4 list dataset fields but not dataset composition. Add sample counts per acquisition mode and robot configuration, including how many samples were removed and why.*

We added the dataset composition (Table S12 and SI Section 4.1): sample counts, acquisition days and sessions, sampled and recorded tendon ranges, logging scheme, and split. For the multi-section arm, each commanded configuration selected one of the three sampling modes of Algorithm S1 with equal probability, so each mode accounts for about one third of the configurations by design; the mode was not stored for each sample. [**FILL: removed batches/samples and reason**]

Sincerely,
Lorenzo Vignoli, Guanran Pei, Francesco Braghin, and Josie Hughes